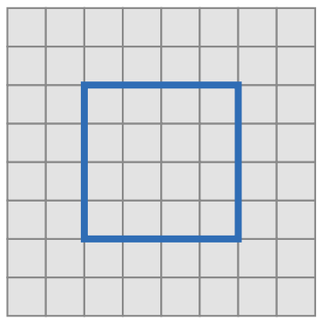
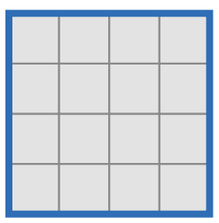
Stage 1
Block Formation

Input image

512 × 512 (gray / color)





4 × 4 blocks
→ 16-D vectors
(N = 16,384)

Color images: RGB channels are processed sequentially; LBG is parallel within each channel

Stage 2A
Sequential LBG (Baseline)

Repeat until K = 64

LBG iteration (k)

1. Nearest-codeword assignment

2. Update codebook (centroids)

3. Convergence check

Codebook C (K = 64)
+ index array L (N = 16,384)

vs.

Stage 2B
MPI-Based Parallel LBG (Proposed)

Equal or Adaptive workload distribution

MPI_Scatterv sends the vectors to the processes

Repeat until K = 64

MPI iteration (k), all processes

Process 0 | Process 1 | ... | Process P − 1

1. Nearest assignment (local vectors)

2. Local sums, counts, distortion

3. MPI_Allreduce (global statistics)

4. Codebook update (all processes)

5. Convergence check

C on all processes; L on Process 0 (MPI_Gatherv)

Stage 3: Lossless Index Coding

Delta coding
offset 128

→

Canonical Huffman Coding
from code lengths

→

Compressed container
header, codebook, code-length table, bitstream

executed by Process 0

Stage 4: Decoding and Reconstruction

Huffman decoding
code-length table and bitstream

→

Inverse delta coding
recovers the VQ indices

→

VQ reconstruction
codebook lookup

executed by Process 0

Parallel: all MPI processes work together

Sequential: a single process